

T3 matched EFT Lagrangian terms: EFT_2_after_S1_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out S1, S2. Remaining active T3 fields: none.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	H	ℓ	e	F	H	ℓ
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0								
										$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$					
										$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$					
cont.										$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}C P_R L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$					
										$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}C P_R L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$					
										$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}C P_R L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$					
cont.										$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}C P_R L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$					
										$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}C P_R L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$					
										$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}C P_R L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$					

Field combinations 2 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	e
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
<i>cont.</i>								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			